Cover letter

14 October 2026

The Editor-in-Chief
Endocrine Connections

Dear Editor-in-Chief,

Please consider “TSHR and IGF1R in Graves disease and thyroid eye disease: genetic evidence from blood expression” as an original research article.

Teprotumumab blocks IGF-1R and is effective in thyroid eye disease (TED). That success raises the question of whether *IGF1R* also carries inherited susceptibility to Graves disease (GD). Therapeutic efficacy and disease susceptibility are different claims, and we asked how consistently genetic evidence links blood expression of *TSHR* and *IGF1R* to disease. Within a Mendelian randomization screen of 2,544 druggable genes, we used Biobank Japan GD for discovery and compared UK Biobank hyperthyroidism and three overlapping FinnGen endpoints: GD, Graves ophthalmopathy and strict Graves ophthalmopathy.

Higher genetically proxied *TSHR* expression was associated with lower GD odds in Biobank Japan ($P = 1.09\times10^{-14}$) and FinnGen GD ($P = 3.19\times10^{-32}$), in the same direction in all five datasets. Colocalization supported a shared causal variant in Biobank Japan and both ophthalmopathy endpoints (in Biobank Japan and the strict endpoint only under the primary prior) but favored distinct variants in UK Biobank and FinnGen GD, the two datasets with the most cases. We report this discordance prominently rather than set it aside: the data establish association, not expression-mediated causality. All *IGF1R* odds ratios exceeded 1, but only two reached nominal significance, and the most precise estimate, in FinnGen GD, was close to the null.

Further analyses test how far these results can be pushed. An allele-direction audit showed that the expression-increasing *TSHR* allele is coupled with the rs179247 allele previously linked to GD protection and greater thymic transcription. Allele-frequency substitution and instrument-omission analyses show where the evidence is robust and where it depends on single variants. A fixed five-tissue GTEx extension and a small orbital RNA-sequencing series (four TED samples, one control) are reported as supplementary and descriptive; the adipose instruments come from the same *TSHR* region and are not presented as independent evidence.

We are equally explicit about what the study does not show. The FinnGen endpoints overlap and use population controls, so they cannot isolate eye-disease susceptibility among people who already have GD. Nothing here predicts treatment response, and weaker genetic evidence for *IGF1R* is not an argument against IGF-1R blockade. The FinnGen extensions and tissue analyses were post hoc and are labeled as such.

The submission includes three main tables, three main figures, a supplement with three tables and two figures, a single results workbook and a STROBE-MR reporting crosswalk.

An earlier, narrower version was considered at thyroid-focused journals; the work was subsequently restructured around the broader genetic screen. The orbital dataset also supports a different, whole-transcriptome question in a separate manuscript under review elsewhere. Here only three genes are described, and no genome-wide expression or pathway catalogue is presented. The companion manuscript can be provided for assessment of overlap.

This manuscript has not been published and is not under consideration elsewhere. Funding, competing interests, ethics approval (Pusan National University Hospital IRB 2104-018-102) and written informed consent are declared in the manuscript.

Sincerely,

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